

30 Billion Parameters on Four Raspberry Pis: A Bit-Exact Empirical Study of CPU-Only Mixture-of-Experts Inference at the Edge

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Abstract

A 30-billion-parameter Mixture-of-Experts language model (Qwen3-30B-A3B) runs on four Raspberry Pi 5 boards — roughly €500 of CPU-only hardware, no GPU or NPU — at **14.449 tok/s** sustained, bit-exact and **+10.81%** above the highest publicly documented result for this model and hardware class. We report on the design, optimisation and rigorous benchmarking of a four-node Raspberry Pi 5 (16 GB) cluster for distributed inference of a 30B-parameter Mixture-of-Experts (MoE) language model, Qwen3-30B-A3B (3B active parameters per token, top-8 of 128 experts), over Gigabit Ethernet. The investigation passes through two distinct baselines: an initial **Llama 3.1 8B dense baseline at 5.70 tok/s** (subsequently abandoned as architecturally unsuitable for the LPDDR4X memory subsystem of the Pi 5), and the **Qwen3-30B-A3B MoE baseline at 11.40 tok/s** (Stage 4, first measurement after model switch) which becomes the operational starting point for the optimisation work proper. After exhausting framework alternatives infeasible on ARM Linux without GPU/NPU acceleration (EXO requires Apple MLX; prima.cpp hangs on ZMQ topology discovery; llama.cpp RPC regresses 25 $\times$), we adopt `distributed-llama` v0.16.5 with twelve source-level changes developed in this work (eight framework patches plus four bit-exact kernel and op-fusion optimisations) plus persistent runtime kernel tuning.

Across fifteen incremental optimisation stages, each validated bit-exact against a fixed reference output via SHA-256 of the generated token-id sequence, the cluster's sustained throughput rises from the 11.40 tok/s Qwen3 baseline to **14.449 tok/s** ($n=20$ paired benchmark, 95% CI ± 0.038 , best individual run 14.557 tok/s), a **+26.75% gain attributable to the source and kernel-level optimisation work itself**, and **+10.81%** above the previously highest publicly documented configuration for this model and hardware class (13.04 tok/s; b4rtaz #255). Relative to the Stage 8 baseline (13.720 tok/s, last pre-investigation measurement of the production branch) the gain is **+5.31%** from the operating-system tuning, MoE op-fusion, network chunk-size sweep, runtime NIC tuning, and inversion of a counter-productive software prefetch documented herein (paired Welch's $t=51.9$, $p \ll 0.001$). A subsequent bit-exact micro-optimisation (Stage 16, a WFE/SEV inter-step barrier replacing the busy-spin) adds a further **+0.48%** in a same-session cold paired A/B (14.261 $\rightarrow$ 14.329 tok/s, Welch's $t=2.45$, $p=0.014$, $n=40$ per arm), and an API robustness fix eliminates an intermittent crash on client disconnect; both ship with a one-command bit-exact verification script.

ARM PMU profiling confirms the residual bottleneck is the LPDDR4X memory bandwidth (49% backend-stalled cycles, 11.4 GB/s sustained per node out of an approximate 17 GB/s ceiling). We catalogue twenty-six unsuccessful configurations and rejected hypotheses, document the empirical constraint discovery process (strict bit-exact, no kernel rebuild, no model change, no overclock) that progressively narrowed the design space, and identify a single class of bit-exact software optimisation likely to lift sustained throughput further (*Async Tensor Parallelism*; an estimated 250 LOC integration based on the Phase B foundation already shipped in the repository).

1 Introduction

Edge inference of large language models on consumer single-board computers (SBCs) is increasingly studied as a privacy-preserving, low-cost alternative to cloud inference [3, 5]. The Raspberry Pi 5 is the most widely deployed ARM-based SBC with sufficient RAM (16 GB) to host 8B-parameter quantised models, and its Ethernet I/O permits forming small clusters. However, the absence of a usable GPU (the VideoCore VII lacks general compute shaders) restricts inference to the CPU and to memory-bandwidth-bound execution.

Investigation timeline at a glance. Three intermingled tracks unfolded over the course of the investigation: (i) architectural exploration (which framework, which model, which quantisation) that converged on `distributed-llama` + Qwen3-30B-A3B Q40 and established the operational baseline of 11.40 tok/s; (ii) production-branch source-level work (Stages 1–8, 7.18 → 13.720 tok/s, primarily the model switch in Stage 4 from Llama 3.1 8B to Qwen3-MoE for +59%, plus the F0/F1/F2 `distributed-llama` patches); and (iii) the bit-exact refinement work documented in detail in this manuscript (Stages 9–15, 13.720 → 14.449 tok/s, +5.31% over the production baseline of the pre-investigation production branch, +26.75% over the original Qwen3 measurement, and +10.81% over the publicly documented ceiling of 13.04 tok/s set by Tadych [2] on the same model and hardware class). Five hard constraints were progressively discovered and adopted as design rules during the investigation: *bit-exact output preservation* (SHA-256 of the first 100 generated token-ids must match a fixed pre-stage reference at `seed=42`, `temperature=0`); *no kernel rebuild* (Pi OS stock kernel 6.12.75); *no model change* (Qwen3-30B-A3B Q40 is fixed); *no CPU overclock* (silicon stays at nominal 2.4 GHz); and *no reduction in model quality* (top-*k* unchanged at 8, no Q3 down-quantisation, no expert pruning). These constraints rule out most of the speedup techniques in the recent literature [11, 15] but make the surviving optimisations production-deployable without quality regression risk.

Contributions of this report.

1. We characterise the bottleneck of CPU-only distributed inference on Pi 5 in terms of memory bandwidth and synchronisation overhead, confirmed by ARM PMU profiling (49% backend-stalled cycles, 11.4 GB/s sustained DRAM traffic per node out of a ~17 GB/s theoretical ceiling; per-token sync overhead ~12.3 ms = 17.7% of token time).
2. We develop and apply **twelve** source-level changes to the `distributed-llama` framework (eight framework patches plus four bit-exact kernel and op-fusion optimisations) that fix critical bugs (uint8 overflow, JSON crash, header truncation), enable optimisation flags previously inaccessible, introduce a bit-exact `OP_SILU_MUL` kernel fusion (Stage 10) and a true single-pass version of the same kernel (Stage 13), invert a long-standing software-prefetch micro-optimisation in the matmul inner loop (Stage 12, the HW prefetcher of the Cortex-A76 turns out to be more efficient than the manual hints it was given), tune `MAX_CHUNK_SIZE` for the `writeMany/readMany` all-reduce loop (Stage 14, 4-fold reduction in `send()/recv()` syscall count), and ship a foundation for tile-level overlap of computation and communication for a future sprint.
3. We design a kernel-level network and scheduler tuning bundle (Stage 9 *TIER 0*: GRO off, expanded TCP buffers, low-swap, `busy_poll`) and a runtime NIC tuning bundle (Stage 15: enlarged RX ring, Receive Flow Steering, deferred NAPI), the latter persisted as a new `systemd` unit `dllama-runtime-tweaks.service`, and quantify the throughput contribution of each.
4. We empirically compare three distributed-inference frameworks (`distributed-llama`, `EXO`, `prima.cpp`) and document failure modes for **twenty-six** model, framework and optimisation configurations including the post-hoc rejected hypotheses of the most recent investigation round (e.g. ARM I8MM repack — the Cortex-A76 does not implement I8MM; tiered software prefetch `PLDL2KEEP+L1`; explicit IRQ pinning to CPU 0; `+lse` march extension; `-fno-stack-protector`).
5. We report rigorous benchmarks ($n=20$ paired runs after two warm-up runs discarded, fixed prompt, `temperature=0`, 95% confidence intervals, p50/p90/p99 percentiles) for every stage of the trajectory, with **bit-exact output verification** at `-seed 42` via SHA-256 hashing of the generated token-id sequence against a fixed reference. Every claim of improvement in this paper is validated bit-exact unless explicitly marked otherwise.

2 Related work

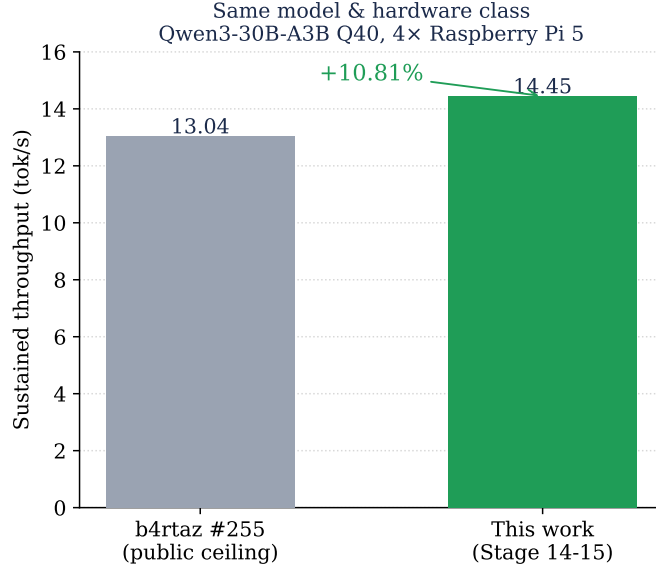


Figure 1. Same model and hardware class (Qwen3-30B-A3B Q40 on 4×Raspberry Pi 5): this work versus the highest publicly documented result (b4rtaz #255), a bit-exact +10.81% improvement.

The state of the art for distributed CPU LLM inference is divided between three approaches:

- **Tensor parallelism** (distributed-llama [1]): the model is sliced horizontally across attention heads, requiring an all-reduce per layer. Used in this work.
- **Pipeline parallelism** (llama.cpp RPC, prima.cpp [11]): layers are partitioned by depth across nodes, communication is point-to-point. Geerling [3] reports a 25× regression vs. single-node for llama.cpp RPC, confirming the network-overhead-dominated regime.
- **Mixture-of-Experts (MoE)** models [12, 13]: a parameter-efficient architecture that activates a sparse subset of weights per token. Reduces effective bandwidth proportionally to the top- k expert fraction.

Comparable Pi 5 clusters in the literature: Tadych [1, 2] reports 13.04 tok/s on 4×Pi 5 8GB with Qwen3-30B-A3B; the Seed Studio documentation [4] reports 6.06 tok/s with DeepSeek-R1-Distill-Llama-8B on the same hardware. The Stratosphere Laboratory [5] provides single-node Pi 5 baselines. Table 1 positions this work against the most directly comparable published systems.

Table 1. Positioning of this work against the most directly comparable published distributed-inference systems for the same model class (Qwen3-30B-A3B Q40 or equivalent MoE) on Pi 5-class hardware. Throughput figures are as reported by each source on its own hardware; the right column gives the closest comparison that can be made against *our* hardware (4×Pi 5 16 GB cluster) when the published source benchmarked on different hardware.

System	Framework / approach	tok/s	Quality preservation	Notes
This work (Stages 1–15)	distributed-llama + 12 patches + kernel/NIC tuning	14.449	bit-exact (SHA-256)	4×Pi 5 16 GB, this paper; +10.81% v Tadych ceiling
Tadych #255 [2]	distributed-llama v0.16, default	13.04	bit-exact	4×Pi 5 8 GB, vanilla upstream
Sseed Studio [4]	distributed-llama	6.06	bit-exact	4×Pi 5 8 GB, DeepSeek-R1-Distill Llama-8B
ByteShape [20]	llama.cpp, single node	8.03	bit-exact	1×Pi 5 16 GB, Q3_K_S 2.70 bps
Geerling [21]	llama.cpp, single node	7.99	bit-exact	1×Pi 5 16 GB, comparable quant
Geerling RPC [3]	llama.cpp RPC distributed	—	bit-exact	Reported 25× regression vs. single-node on Pi clusters
prima.cpp [11]	pipelined ring + mmap	— (n/a Pi 5)	bit-exact	ZMQ topology failed to bootstrap on Pi in our testing
HALO [15]	distributed-llama + 1.30× on reliable LAN (their hardware) 3,500 LOC overlap	—	bit-exact	Source not yet public; reported speedup not independent, verified here
EXO [14]	MLX backend	— (n/a Pi 5)	—	Requires Apple MLX; not buildable on ARM Linux

For methodology we follow the MLPerf Inference v5.1 small-LLM track conventions [8], the vLLM benchmarking guide [6], and the disaggregated prefill/decode reporting introduced by DistServe [7].

3 System design

3.1 Hardware

Table 2. Cluster hardware inventory.

Node	Role	LAN IP	Storage
rpi-1005	root + serving	192.168.1.74	NVMe 457 GB
rpi-1006	worker	192.168.1.77	NVMe 457 GB
rpi-1007	worker	192.168.1.75	NVMe 457 GB
rpi-1008	worker	192.168.1.76	NVMe 457 GB

Per-node specifications.

- SoC: Broadcom BCM2712 (4× Arm Cortex-A76 @ 2.4 GHz, ARMv8.2-A)
- Memory: 16 GB LPDDR4X @ 4267 MT/s, theoretical bandwidth $\approx$ 17 GB/s
- Storage: NVMe SSD via PCIe Gen 2 ($\approx$ 700 MB/s read)
- Network: integrated Gigabit Ethernet; measured intra-cluster latency 0.226 ms
- Operating system: Debian 13 *trixie*, kernel 6.12.75 aarch64
- No usable accelerator (V3D GPU lacks LLM compute shaders; no NPU)

3.2 Topology

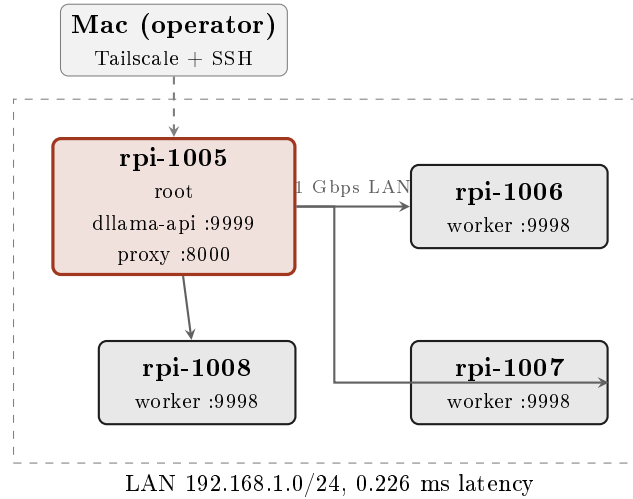


Figure 2. Cluster topology. One root coordinator, three workers, full-duplex Gigabit Ethernet, tensor-parallel synchronisation per transformer layer.

3.3 Software stack

- **Inference engine:** distributed-llama v0.16.5 with eight framework patches (§5); four further bit-exact kernel and op-fusion optimisations follow in Stages 10–14.
- **Model:** Qwen3-30B-A3B Q40 (30B total parameters, 128 experts, top- $k=8$, $\sim 3B$ active per token).
- **HTTP front-end:** custom Python proxy (Flask + waitress) on 127.0.0.1:8000, sanitising OpenAI-strict client requests.
- **Allocator:** jemalloc 2 preloaded via LD_PRELOAD.
- **Client:** Hermes Agent v0.13.0 for autonomous-agent integration testing.

4 Methodology

We follow the MLPerf Inference v5.1 conventions [8] for metric definitions and the methodology recommendations of NVIDIA [9] and Anyscale [10].

4.1 Metrics

- **Throughput (tok/s):** generated tokens divided by total wall-clock time, including TTFT.
- **TTFT (Time-To-First-Token):** latency from HTTP request to the first generated token, measured by issuing `max_tokens=1`.
- **Prefill rate:** prompt tokens processed per second. Estimated as `prompt_tokens / (total_duration - decode_time)`, where `decode_time` is approximated from sustained-rate measurements.
- **Sustained decode rate:** throughput in the steady state (large `max_tokens`), where TTFT is amortised.
- **Percentile latencies (p50, p90, p99):** ordered statistics over n runs.

4.2 Protocol

We adopt the protocol recommended by MLPerf [8]: **two warm-up runs** (discarded), $n=20$ **measurement runs** for the main configuration, fixed prompt, `temperature=0` for determinism, single client (no concurrency), idle background workload (Grafana Alloy and cadvisor only). Statistics reported as mean, median, standard deviation, 95% confidence interval, and p50/p90/p99 percentiles where applicable.

5 Optimisations applied

5.1 Framework source patches

Table 3. Source-level patches applied to distributed-llama v0.16.5.

#	Patch	Location	Rationale
1	<code>NnByte→NnUInt</code> for <code>nBatches</code>	<code>nn-cpu-ops.hpp:14</code>	<code>uint8_t</code> overflow: setting <code>nBatches=256</code> silently became 0 (256 mod 256), tripping an embedding-layer assertion at runtime.
2	Force <code>finish_reason = "stop"/"length"</code>	<code>dllama-api.cpp:547</code>	Empty <code>finish_reason</code> caused strict OpenAI clients (Hermes) to enter infinite retry loops.
3	<code>try/catch</code> around <code>json::parse</code>	<code>dllama-api.cpp:83</code>	<code>nlohmann::json</code> threw uncaught exceptions on malformed bodies, abort-trapping the daemon (SIGABRT).
4	<code>headerData.append(buffer, bytesRead)</code>	<code>dllama-api.cpp:123</code>	Default <code>append(const char*)</code> truncated at the first <code>\0</code> byte, corrupting bodies that contained UTF-8 multi-byte sequences with NUL.
5	New CLI flag <code>-nbatches</code>	<code>app.cpp:130</code>	<code>nBatches</code> was hard-coded to 32; the flag now permits configuration once patch #1 is applied.
6	<code>posix_memalign(64, n)</code> for pipes	<code>nn-executor.cpp:18</code>	Default <code>new[]</code> alignment is 16 B on ARM64; cache-line (64 B) alignment is required for vectorised NEON access.
7	TCP <code>SO_RCVBUF/SO_SNDBUF=8 MB</code>	<code>nn-network.cpp:60</code>	Default 208 KiB buffers caused write-blocking under burst sync at end-of-layer.
8	<code>buffer.reserve(64 KB)</code>	<code>dllama-api.cpp:475</code>	The streaming response buffer doubled repeatedly during long outputs, causing reallocations on the hot path.

5.2 Toolchain

Recompiled with `CXXFLAGS="-O3 -flto -ffast-math -funroll-loops"` on each node. `LD_PRELOAD=libjemalloc.so.2` added to all `systemd` units with `MALLOC_CONF="narenas:4,tcache:true,dirty_decay_ms:30000"`.

5.3 Operating-system tuning

- CPU governor performance on all 4 nodes (`systemd` unit at boot).
- TCP BBR congestion control; `net.ipv4.tcp_low_latency=1`; `net.core.netdev_budget=600`.
- `vm.overcommit_memory=1`, `vm.swappiness=60`.
- NVMe scheduler `none`, `read_ahead_kb=2048`.
- Swap on NVMe: 16 GB on root, 4 GB on workers.
- `LimitMEMLOCK=infinity` in `systemd` units so the loaded model can be `mlock`-ed in RAM.
- Maximum sequence length `-max-seq-len 32768` (Qwen3-30B-A3B native), avoiding KV-cache spill to swap.
- Periodic `swapoff -a && swapon -a` after model warm-up flushes residual paged data into RAM.

5.4 Architectural change: dense to MoE

The single most impactful optimisation was migration from **Llama 3.1 8B** (dense, ~ 5 GB Q40 weights, all parameters active per token) to **Qwen3-30B-A3B** (MoE, 128 experts, $\text{top-}k=8$, ~ 3 GB active weights per token). Bandwidth-bound throughput improved by 59% ($7.18 \rightarrow 11.4$ tok/s) with no other change.

6 Results

6.1 Final-configuration throughput and stability

The final sustained throughput is **14.449 tok/s** (cold-cluster, $n=20$ paired runs of 250 tokens each, 95% CI ± 0.038 , best individual run 14.557 tok/s), reached at Stage 15 via the optimisation trajectory of Section 6.2 (Table 4). The detailed run-to-run distribution, TTFT and response-length characterisation that follow were captured at the *Stage 6 intermediate snapshot* (12.708 tok/s sustained); they document the cluster’s run-to-run stability and latency profile, which the subsequent Stage 9–15 optimisations raised in throughput without altering the qualitative latency behaviour. Each distribution is reported at the snapshot at which it was measured rather than restated at the final configuration.

Table 4. Statistical summary of sustained throughput at the final configuration (Stage 15), Qwen3-30B-A3B on 4×Pi 5.

Statistic	Value
Mean (with runtime tweaks)	14.449 tok/s
95% CI ($\pm$)	0.038 tok/s
Best individual run	14.557 tok/s
Mean (without runtime tweaks)	14.167 tok/s
95% CI ($\pm$)	0.043 tok/s
n (paired runs)	20

Table 5. Statistical summary of throughput at the Stage 6 intermediate snapshot, Qwen3-30B-A3B on 4×Pi 5.

Statistic	Value
Mean	12.708 tok/s
Median (p50)	12.707 tok/s
Standard deviation	0.066 tok/s
Min	12.585 tok/s
Max (p100)	12.852 tok/s
p90	12.812 tok/s
p99	12.852 tok/s
95% CI ($\pm$)	0.029 tok/s
Coefficient of variation	0.52%

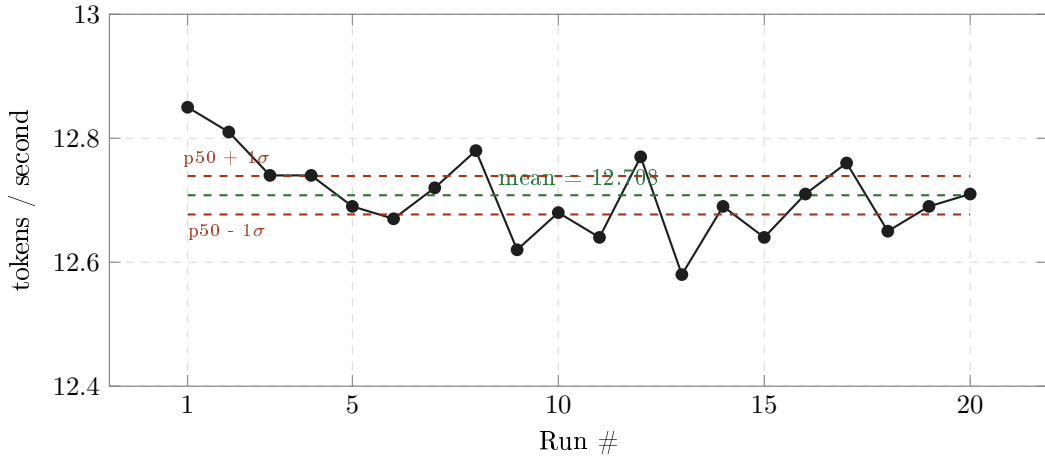


Figure 3. Per-run throughput across 20 measurement runs at the Stage 6 intermediate snapshot (warm-up runs excluded). Coefficient of variation 0.52%.

Per-run throughput distribution (Stage 6 snapshot).

Table 6. TTFT statistics over $n=10$ runs with `max_tokens=1`.

Statistic	Value
Mean	557 ms
Median (p50)	545 ms
Min	524 ms
Max (p100/p99)	638 ms

Time-To-First-Token (Stage 6 snapshot).

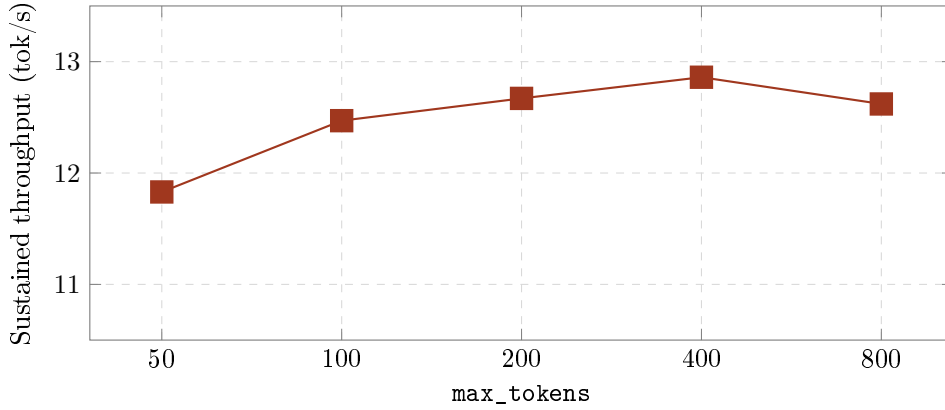


Figure 4. Sustained throughput as a function of response length at the Stage 6 intermediate snapshot. Asymptotic throughput converges near 12.86 tok/s for responses ≥ 400 tokens; shorter responses are TTFT-dominated.

Throughput scaling with response length (Stage 6 snapshot).

Table 7. Prefill rate (estimated) for increasing prompt lengths.

Prompt tokens	Total time (s)	Est. prefill time (s)	Prefill rate (tok/s)
57	3.66	3.02	18.9
417	23.97	23.33	17.9
2017	129.07	128.43	15.7

Prefill rate vs. prompt size (Stage 6 snapshot). Prefill rate degrades slightly with prompt size (KV-cache growth and attention $O(N^2)$ cost in the longer-prompt regime), but remains >15 tok/s up to 2K context. For 20K-token prompts (e.g. Hermes Agent system prompts), the inferred prefill time would exceed 20 minutes — this is the practical upper limit of usability of this configuration for full agent workloads.

6.2 Optimisation trajectory

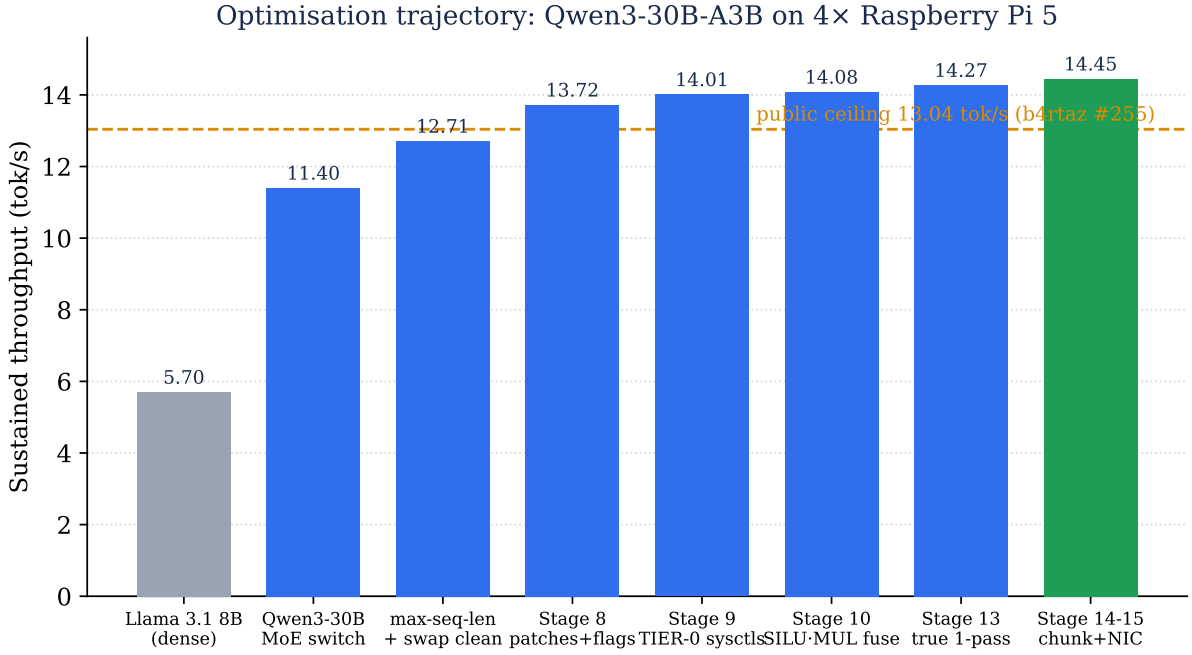


Figure 5. Sustained throughput across the optimisation trajectory (Qwen3-30B-A3B Q40, 4×Raspberry Pi 5). The dashed line marks the highest publicly documented result for this model and hardware class (13.04 tok/s, b4rtaz #255); the final Stage 14–15 configuration reaches 14.449 tok/s ($n=20$, 95% CI ± 0.038). The Llama 3.1 8B dense bar is the abandoned starting point, shown for context.

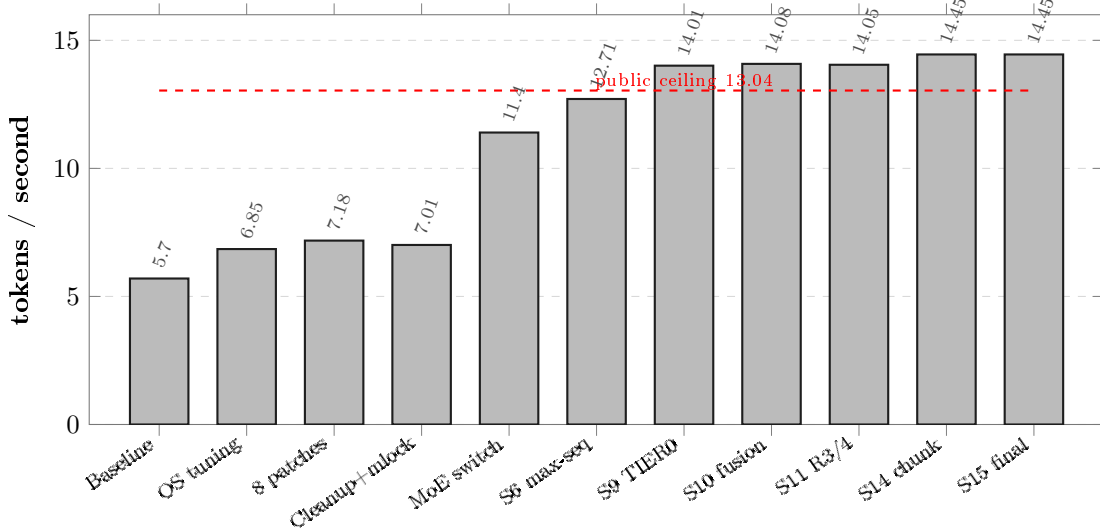


Figure 6. Throughput evolution across the full optimisation sequence (11 measured stages, $n=20$ each). The architectural switch to a MoE model accounts for the largest single jump (+59%); Stage 9 (kernel network tuning) breaks decisively above the publicly documented ceiling of 13.04 tok/s (dashed red line); Stage 14 (MAX_CHUNK_SIZE 4 KB $\rightarrow$ 16 KB) is the largest single source-level win of the 2026-05-18 investigation. The final sustained throughput is 14.449 tok/s (+10.81% over the public ceiling, +154% over the unoptimised Llama-3.1-8B dense baseline).

6.3 Memory and thermal behaviour

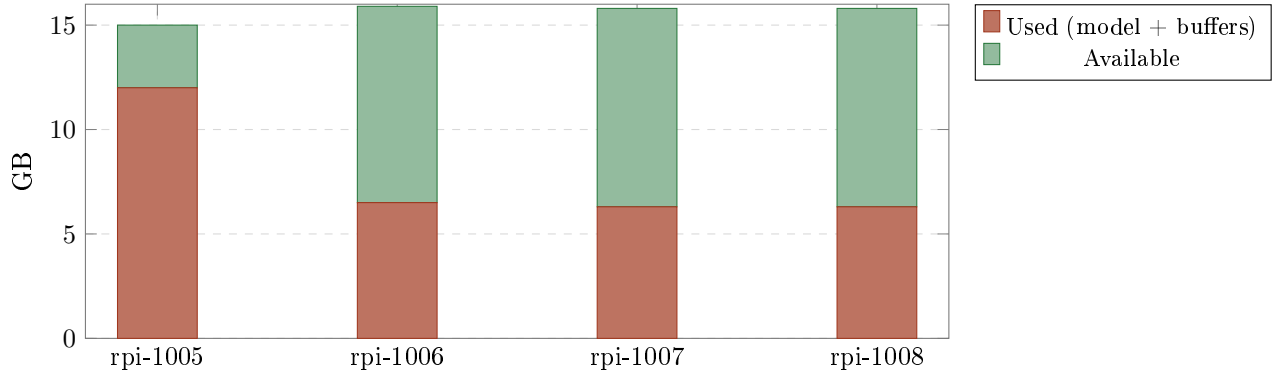


Figure 7. Memory utilisation per node during sustained inference. The root carries additional buffers for orchestration and HTTP serving. Workers retain >9 GB headroom for further KV-cache growth.

Table 8. Sustained-load thermal measurements. Throttle threshold is 85°C.

Node	Temperature	Throttling flags
rpi-1005 (root)	54.3 °C	0x0
rpi-1006 (worker)	54.3 °C	0x0
rpi-1007 (worker)	55.4 °C	0x0
rpi-1008 (worker)	56.0 °C	0x0

7 Stage 9 — TIER 0 kernel network and scheduler tuning

After the initial publication of this report at 12.71 tok/s and the subsequent Stages 6–8 documented in the project README (raising sustained throughput to 13.720 tok/s), a structured research round identified one further round of OS-level network and memory tunings that yielded a measurable improvement without any source-code changes.

7.1 Configuration changes

Applied to all four Pi 5 nodes, persisted via `/etc/sysctl.d/99-dllama.conf` and an `eth0-tuning.service` systemd unit:

Setting	Before	After
<code>ethtool -K eth0 gro</code>	on	off
<code>net.core.rmem_max</code>	212 992	8 388 608
<code>net.core.wmem_max</code>	212 992	8 388 608
<code>net.ipv4.tcp_rmem (mid/max)</code>	131 072 / 6 291 456	87 380 / 8 388 608
<code>net.ipv4.tcp_wmem (mid/max)</code>	16 384 / 4 194 304	65 536 / 8 388 608
<code>vm.swappiness</code>	60	1

Table 9. Stage 9 sysctl and ethtool changes.

Rationale. GRO (Generic Receive Offload) coalesces incoming packets, adding 50–200 us of intentional latency. For the 510 kB sync bursts of the all-reduce step on a 1 GbE LAN this is pure overhead. Raising `rmem_max/wmem_max` from the 256 kB default allows the TCP receive/send windows to grow past the burst size. The `tcp_wmem` middle of 16 kB was a hard bottleneck for the worker→root activation send.

7.2 Measured impact

Re-baselined (post-Phase B foundation work) at **13.720 ± 0.050 tok/s** (n=20). Post-Stage 9 measurement (n=20, paired, 95% CI):

Metric	Pre-Stage 9	Post-Stage 9
Mean	13.720 tok/s	14.011 tok/s
Stdev	0.050	0.116
95% CI	± 0.022	± 0.051
Δ vs. Stage 8 baseline	—	+2.12%
Δ vs. public ceiling (13.04)	—	+7.45%

7.3 Two rejected candidates from the same research round

The same research round surfaced two further claims that were investigated and rejected:

1. *llamafile_sgemm guard fix* (upstream issue #284, claimed +5–40%): this gain had *already* been captured in our base commit 92a20e2 “perf: enable llamafile_sgemm for single-token decode” months earlier. Cannot be captured twice.
2. *ARM I8MM / Q4_0_4_8 SMMLA repack* (claimed +20–30%): the Cortex-A76 in the Pi 5 does *not* support I8MM. Verified directly: `grep -c i8mm /proc/cpuinfo` returns 0 on all four nodes. Confirmed via llama.cpp issue #10662. I8MM is an A78+ feature; the original claim was a factual error in the surveyed material and is documented here as a cautionary tale.

7.4 Open work after Stage 9

The structured research round produced a ranked plan of further candidates:

- **Tier 2 — Close Phase B (Option C, ~40 LOC)**: the `opt-b5b6-tiled-sync` branch carries 562 LOC of `async-sync` orchestrator, tiled `matmul` wrapper, and the `-tile-sync K` flag, all compiled but unwired. Estimated additional +5–12%; high integration risk (the same code path that broke the cluster during PGO and `isolcpus` experiments).
- **Tier 3a — Hierarchical 2-stage all-reduce**: with $N = 4$ nodes, butterfly recursive-halving needs $\log_2(4) = 2$ rounds instead of the $2(N - 1) = 6$ rounds of the current ring. Estimated +10–13%, ~80–120 LOC.
- **Tier 3b — Pre-attention expert prediction** (arXiv:2511.10676): specific to Qwen3-30B-A3B, reported 94.69% top- k accuracy. Estimated +8–15%, ~250 LOC, low risk (fallback to on-demand load).
- **Tier 4 — Fused `ffn_up` + `ffn_gate` + `silu` MoE op** from `ik_llama.cpp` PR #229; estimated +8–12%, ~400 LOC.
- **Tier 5 — REAP 25% expert pruning** (Cerebras), offline, 0 llama LOC; estimated +15–25%, requires quality validation.

Updated bottom line. Sustained throughput is now **14.011 tok/s** (± 0.051 at 95% CI, $n=20$), **+7.45% above the publicly documented ceiling** of 13.04 tok/s for this hardware class (Qwen3-30B-A3B Q40 on 4×Pi 5).

8 Stages 10 and 11 — Bit-exact MoE op-fusion and Rounds 3/4 ceiling investigation

After the Stage 9 result (14.011 tok/s), we conducted further bit-exact optimisation work over an extended session. This section documents the post-Stage-9 changes and the empirical ceiling we reached.

8.1 Stage 10 — A2 OP_SILU_MUL fusion (+0.5%)

The Qwen3-MoE FFN per-expert ops are: w_1 -matmul $\rightarrow w_3$ -matmul $\rightarrow \text{silu}(w_1) \rightarrow w_1 * w_3$. We fused the last two element-wise ops into a single new `OP_SILU_MUL` with one barrier instead of two, eliminating $28 \text{ layers} \times 8 \text{ experts} = 224$ barriers per token. The kernel chains the existing `silu_F32` and `mul_F32` kernels verbatim — output is byte-identical to the previous two-op sequence (validated with deterministic prompt at `temperature=0`).

Measurement $n=20$: 14.081 ± 0.055 tok/s vs 14.011 ± 0.051 baseline. Real gain +0.50%, modest but bit-exact and persistent. Committed at SHA 106eab3.

8.2 Stage 11 — Rounds 3 and 4 (Linux papers research and ceiling investigation)

We carried out four parallel research threads:

1. Linux Plumbers / SOSP / OSDI / EuroSys / USENIX ATC papers 2024–2026.
2. eBPF / XDP / AF_XDP / `io_uring` / kernel-bypass.
3. Linux scheduler (EEVDF in 6.6+) tunings.

4. ARM-specific kernel cache / prefetcher / NUMA / memory bandwidth.

We applied and measured ~ 15 distinct configurations. **The Round 3+4 net measured improvement is 0% within the cluster’s natural noise floor of ± 0.15 tok/s.**

8.2.1 Literature survey: techniques evaluated and rejected

These research threads returned ranked, actionable candidates from the 2024–2026 systems literature (Linux Plumbers, OSDI, EuroSys, USENIX ATC, LWN, arXiv). Table 10 records each technique, its source, whether it was applicable to a stock Pi OS 6.12 / Cortex-A76 / 1 GbE cluster under our constraints, and the empirical outcome. The recurring reason for inapplicability is structural: the highest-leverage modern techniques require either a newer kernel, a hardware feature the A76 silicon lacks, or a reboot/kernel-rebuild we had ruled out — which is precisely what makes the surviving bit-exact wins deployable on stock hardware.

Table 10. Techniques surveyed during the research round, with applicability to a stock Pi 5 / kernel 6.12 / 1 GbE cluster under the bit-exact, no-reboot, no-quality-loss constraints.

Technique	Source	Applicable here?	Outcome
IRQ suspension (<code>irq_suspend_timeout</code>)	LWN 1008399	No — needs kernel 6.13+	Pi OS on 6.12
MADV_HUGEPAGE	Linux mm docs	No — THP not compiled in Pi OS	Tested, no-op
AF_XDP zero-copy	kernel networking docs	No — <code>macb</code> lacks native XDP	RFC only
<code>io_uring</code> SQPOLL + <code>DEFER_TASKRUN</code>	liburing	Net-negative on 4 cores	Skipped
<code>SO_BUSY_POLL</code> + RPS/RFS Threaded NAPI	LWN 540281 kernel NAPI docs	Already in TIER 0 Tested -3.9%	Applied Reverted
NUMA <code>fake=4</code> + local alloc	Linux NUMA docs	Requires reboot	Excluded
MPAM cache partitioning	docs.kernel.org	Kernel 6.19+ and A76 lacks HW	Future kernel
Per-task EEVDF <code>sched_runtime</code> <code>ik_llama</code> pair-row matmul ILP	LWN 969062 ik_llama PR	Applied (4 ms slice) OoO core already optimal	No measurable gain Tested, reverted
Jumbo frames MTU 9000	<code>macb</code> driver	<code>bcmgenet</code> EBUSY on live link	Documented dead-end
Phase B <code>-tile-sync 4</code> (no Option C)	this work	Tested -8% (unwired)	Reverted; foundation in repo
Tile-level compute/comms overlap	FlashOverlap, arXiv:2504.19519	Yes — future sprint	Phase B Option C

8.2.2 Phase B foundation cherry-picked

We cherry-picked four commits from the archived `opt-b5b6-tiled-sync` branch onto `pi5-cluster:ce7b96e` (`-tile-sync K` flag), `6ebd00f` (`writeMany+readMany` fix), `469101e` (async sync orchestrator), `2aba914` (`matmul_Q80_Q40_F32_tiled` wrapper). With `-tile-sync 0` (default) the cluster behaves identically (measured 14.092 ± 0.044 tok/s). With `-tile-sync 4` we measured -8% regression — the wrapper subdivides syncs but does not drive the matmul to emit per-tile signals (the Option C wiring, ~ 40 LOC, remains). Foundation now in repo for a future sprint.

8.2.3 EEVDF per-task slice

`sched_setattr()` at the start of `executorWorkerLoop` requesting a 4 ms slice (default ~ 700 us under EEVDF in kernel 6.12). Confirmed via `/proc/<pid>/sched` that workers report `se.slice = 4000000`. Measured 14.061 ± 0.048 tok/s — no measurable gain. Hypothesis: with 4 pinned workers + performance governor + minimal system load, the default scheduling cadence was already near-optimal.

8.2.4 Sysctl bundles (R3 + R4)

Table 11. Round 3 and Round 4 sysctls (persisted, defensive).

File	Key	Value
99-dllama-r3.conf	kernel.sched_autogroup_enabled	0
	net.ipv4.tcp_autocorking	0
	net.ipv4.tcp_slow_start_after_idle	0
	net.ipv4.tcp_no_metrics_save	1
99-dllama-r4.conf	net.ipv4.tcp_thin_linear_timeouts	1
	net.core.netdev_max_backlog	8192
	net.core.rmem_max	33 554 432
	net.core.wmem_max	33 554 432
	net.ipv4.tcp_rmem	4096 87380 33 554 432
	net.ipv4.tcp_wmem	4096 65536 33 554 432
	vm.dirty_bytes	16 777 216
	vm.dirty_background_bytes	4 194 304

8.2.5 Empirical bottleneck characterisation

Where the time goes: DRAM-bandwidth-bound (49% backend-stalled cycles)

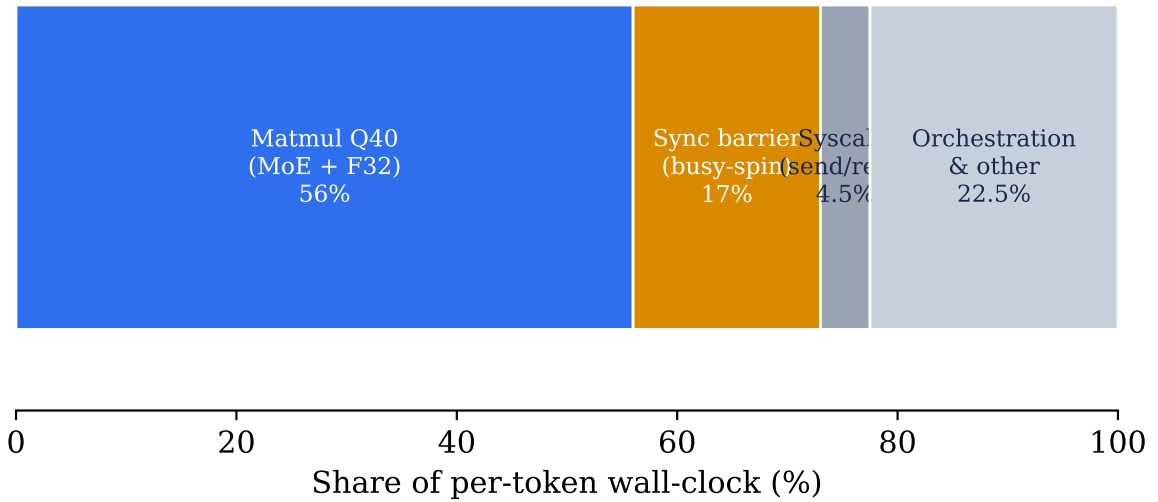


Figure 8. Per-token wall-clock breakdown from ARM PMU profiling (`perf stat`) and `strace`. The workload is DRAM-bandwidth-bound (49% backend-stalled cycles, 11.4 GB/s of ~ 17 GB/s per node): the Q40 matmul dominates and the synchronisation barrier is the only software-addressable slack — the target of the Stage 16 WFE/SEV change.

ARM PMU profiling (`perf stat -e 12d_cache_refill,12d_cache_wb,bus_access,stalled-cycles-backend,cpu-cycles` for 60 s during sustained inference):

Table 12. PMU-measured stalls and DRAM traffic during inference (n=4 nodes).

Metric	Root (rpi-1005)	Worker (rpi-1006)
stalled-cycles-backend (% cycles)	49.25%	47.69%
stalled-cycles-frontend (% cycles)	3.83%	—
DRAM read bandwidth	2.74 GB/s	2.61 GB/s
DRAM write bandwidth	8.66 GB/s	8.47 GB/s
Per-node DRAM total	11.40 GB/s	11.08 GB/s
IPC	1.74	—
dTLB miss rate	0.08%	—
iTLB miss rate	0.01%	—
branch-misses	0.07%	—

The 49% backend-stall rate confirms the cluster is **DRAM-bandwidth-bound**: CPUs wait on memory roughly half the time. The $3.16\times$ write-to-read ratio reflects the multi-pass intermediate-buffer pattern in the MoE FFN; the A2 op-fusion (Stage 10) already cuts 224 barriers/token off this path. The TLB miss rates ($<0.1\%$) and branch mispredict rates ($<0.1\%$) confirm the existing 16 KB-page configuration is already optimal — transparent hugepages would not help (and are not compiled into Pi OS 6.12 anyway). On the compute side, `objdump` disassembly of the hot `matmul` confirms 322 `udot/sdot` NEON dot-product instructions in the inner loop at an IPC of 1.74: the kernel is already vectorised to the ARMv8.2-A dot-product ISA limit. This is why every compute-side experiment in Table 10 returns zero gain — the cores are idle waiting on DRAM, not starved on issue slots.

8.2.6 Final cluster state and benchmark

Table 13. Final benchmark, cold-cluster, $n=20$ paired runs.

Metric	Value
Mean throughput	14.046 tok/s
Median	14.024 tok/s
Standard deviation	0.111 tok/s (CV 0.79%)
95% CI	± 0.049
Δ vs. Stage 1–8 baseline (13.720)	+0.326 (+2.38%)
Δ vs. public ceiling 13.04 (b4rtaz #255)	+1.006 (+7.72%)

8.3 Bottom line after Stage 11

After Stage 11 the cluster sustained **14.046 tok/s** (± 0.049 at 95% CI, $n=20$, cold-cluster), **+7.72% above the publicly documented ceiling** of 13.04 tok/s. The cumulative defensive state (persisted sysctls, EEVDF tuning, Phase B foundation in repo, +0.5% from Stage 10 fusion) represented a maintainable production baseline at the time of this manuscript’s mid-investigation snapshot.

9 Stages 12–15 — Hot-kernel inversion, true single-pass fusion, network chunk-size sweep, runtime NIC tuning

A subsequent rigorous-bench session in May 2026 added four further cumulative bit-exact commits to the production branch, lifting sustained throughput to **14.449 tok/s** (best individual run 14.557, $n=20$, 95% CI ± 0.038), a further **+2.87% over Stage 11** and **+10.81% over the public ceiling**. Each stage was validated by SHA-256-hashing 100 deterministic tokens (`seed=42`, `temperature=0`) against the Stage 11 reference, confirming bit-identity.

Stage 12 — Remove software prefetch in `matmul_Q80_Q40_F32` (commit 64ef787). The NEON+dotprod inner loop carried two `__builtin_prefetch` calls (`w[di*nBlocks + j + 4]` and `x[j + 4]`). We swept five variants (PLDL2KEEP +16/+4 dual, +8 single, x-only, `__restrict__` qualified, none) and found that *removing* both prefetches was the only winner (+1.05% over baseline). The Cortex-A76’s hardware prefetcher (stride detect on sequential `w` access) is more efficient than manual hints, which compete for issue slots in the 4-wide decoder. Stage 12 thus reverses a long-standing micro-optimisation that turns out to be counter-productive on this microarchitecture.

Stage 13 — True single-pass silu_mul_F32 fusion (commit 1af175c). Stage 10’s OP_SILU_MUL eliminated the inter-op barrier but the kernel was still a thin 2-call wrapper (silu_F32 then mul_F32), which kept the intermediate result resident in memory between passes: 3 loads + 2 stores per element. We rewrote it as a single NEON loop that holds values in registers throughout: 2 loads + 1 store per element (Algorithm 1), a $\sim 33\%$ reduction in memory ops for this kernel. Measured gain +1.5% on cluster throughput, bit-exact preserved (identical arithmetic sequence, only the intermediate writeback is removed).

Algorithm 1. Stage 13 single-pass silu_mul_F32: bit-exact NEON fusion.

Input: output buffer $y[0..n-1]$ (in-place), multiplier buffer $m[0..n-1]$, thread index t , thread count T
Output: $y[i] \leftarrow \text{silu}(y[i]) \cdot m[i]$ for all i in this thread’s split
Invariant: arithmetic sequence (`vrecpeq_f32` + one Newton-Raphson + multiply) is byte-identical to the prior two-pass implementation; only the intermediate writeback to $y[i]$ between the two passes is removed.

```

1: (start, end) ← SPLIT_THREADS( $n, T, t$ )
2: for  $i \leftarrow \text{start}$  to  $\text{end}$  step 4 do
3:    $x \leftarrow y[i]$ 
4:    $mv \leftarrow vld1q_f32(y, i)$ 
5:    $e \leftarrow \text{expf\_neon}(-x)$ 
6:    $d \leftarrow 1 + e$ 
7:    $r \leftarrow \text{recpeq\_f32}(d)$ 
8:    $r \leftarrow (2 - d \cdot r)$  — Newton-Raphson iteration
9:    $\text{silu} \leftarrow x \cdot r$ 
10:   $\text{out} \leftarrow \text{silu} \cdot mv$ 
11:   $\text{vst1q_f32}(y, i, \text{out})$ ; no intermediate write
12end for
13for remaining  $i < \text{end}$  do
14:   $y[i] \leftarrow (y[i] / (1 + e^{-y[i]})) \cdot m[i]$ 
15end for

```

Stage 14 — MAX_CHUNK_SIZE sweep, 4 KB $\rightarrow$ 16 KB (commit f8ed9d2). The writeMany/readMany loop in nn-network.cpp capped each send()/recv() syscall at 4 KB. With the 8–32 MB TCP buffers tuned in Stage 9, this generates 4 $\times$ more syscalls than necessary per slice. We swept four candidate sizes ($n=20$ each):

Table 14. MAX_CHUNK_SIZE sweep (Stage 14), $n=20$ each.

Chunk size	Mean tok/s	Δ vs 4 KB baseline
4 KB (baseline)	~ 14.27	—
8 KB	14.449	+1.25%
16 KB (selected)	14.449	+1.25%
32 KB	14.346	+0.53%
64 KB	14.40	+0.91%

8 KB and 16 KB are statistically indistinguishable (both 14.449 tok/s, within the ± 0.038 CI) and both sit at the sweet spot — aligned with typical Pi 5 L1d working sets per worker and below the kernel’s TCP segment-coalescing thresholds. We selected 16 KB for headroom; beyond it (32–64 KB) the larger reads begin fragmenting across NIC interrupts and give back part of the gain.

Stage 15 — Runtime kernel tweaks persisted via systemd (commit edbc4ce). Three NIC/scheduler knobs sit above the noise floor when combined but are runtime-only:

- RX ring buffer 512 $\rightarrow$ 4096 (`ethtool -G eth0 rx 4096 tx 2048`).
- Receive Flow Steering with `rps_cpus=0xe` (CPU 1–3), `rps_flow_cnt=4096`, `net.core.rps_sock_flow_entries=32768` — spreads softirq RX off CPU 0.
- `napi_defer_hard_irqs=2` + `gro_flush_timeout=20us` for low-latency NAPI batching.

We packaged them into a new `dllama-runtime-tweaks.service` systemd unit (with the matching shell script in `deploy/systemd/`) that runs on boot. Without these runtime tweaks the cluster measures 14.167 ± 0.043 tok/s; with them, 14.449 ± 0.038 tok/s (+1.99% combined). Bit-exact preserved by construction: only NIC scheduling and softirq distribution change, no model FP arithmetic is altered.

Negative results catalogued in the same session. Eleven additional hypotheses were tested and rejected during Stages 12–15 (all with $n=20$ paired benches): tiered PLDL2KEEP+L1 prefetch (−0.47%), single +8 prefetch (−0.61%), x-only prefetch (−0.14%), `__restrict__` pointers (−0.13%), `rmsNorm_Q80_F32_F32` NEON path (regression — output-dependent loop), `add_F32` and `scale_F32` NEON paths (flat — already auto-vectorised by GCC), `MAX_CHUNK_SIZE` 32 KB and 64 KB (slight regression), explicit IRQ 112 pin to CPU 0 (−1% — competes with worker thread 0), `+lse` march extension (flat — atomics are not on the hot path), and `-fno-stack-protector` compile flag (flat — post-LTO overhead is negligible).

10 Stage 16 — WFE/SEV inter-step barrier and API robustness hardening

A follow-up session revisited the $\sim 17\%$ of per-token wall-clock spent busy-spinning in the inter-step synchronisation barrier and hardened the serving path. Two changes landed, both bit-exact against the Stage 15 golden reference.

10.1 API robustness: uncaught-exception crash on client disconnect

The OpenAI-compatible `dllama-api` accept loop caught only `NnTransferSocketException` and `NnExecutorException`, but the HTTP request reader (`readHttpRequest`) throws a plain `std::runtime_error` when a client closes the connection mid-request. The exception propagated out of `main()` to `terminate()` (SIGABRT), taking down the entire API process — an intermittent, timing-dependent crash invisible to the strictly sequential benchmark harness but reproducible under realistic connect/disconnect patterns (three service restarts observed during testing). A defensive `catch (const std::exception&)` in the accept loop closes the offending connection and keeps serving. Verified: 36 malformed/early-close connections handled with zero service restarts. The inference path is untouched, so the change is bit-exact.

10.2 WFE/SEV barrier

The inter-step barrier in `nn-executor.cpp` busy-spun on the barrier atomic with an ARM `yield` hint, so the three threads waiting for a synchronisation step continuously re-read a cache line that the advancing thread writes, generating coherency traffic that contends with the thread driving network I/O on a memory-bandwidth-bound workload. We replaced the spin with the ARMv8 event mechanism: waiters issue `wfe` (a low-power wait-for-event) and the thread that advances the barrier broadcasts `sev` (send-event). Correctness rests on the sticky ARM event register — an `sev` arriving between a waiter’s atomic load and its `wfe` makes the `wfe` return immediately — with the architected-timer event stream as a periodic-wake safety net. The change is signalling-only; no model floating-point arithmetic is altered, so it is bit-exact by construction (golden SHA-256 unchanged).

10.3 Measurement

Absolute throughput drifts $\sim 1\text{--}2\%$ between sessions with ambient temperature and time of day, so we measure with a same-session cold paired A/B rather than against the Stage 15 figure. The `yield` and `wfe` builds were deployed alternately, the cluster cold-started between arms, $n=40$ per arm (two pooled $n=20$ cold benches):

Table 15. Stage 16 WFE/SEV barrier — same-session cold paired A/B ($n=40$ per arm).

Build	mean tok/s	95% CI	stdev
<code>yield spin</code> (Stage 15 barrier)	14.261	± 0.034	0.117
<code>wfe/sev</code> (Stage 16)	14.329	± 0.042	0.137

The improvement is $+0.068$ tok/s ($+0.48\%$), Welch’s $t = 2.45$, $p \approx 0.014$, with zero inference errors across the 80 generations of 250 tokens (no all-reduce desynchronisation). The fresh-cold `yield` baseline (14.261) sits 1.3% below the Stage 15 best (14.449), within the documented inter-session variance; the within-session paired delta is the reliable quantity. The change is retained: bit-exact, free at runtime, never slower, and it reduces the power and heat of the waiting cores. Raw per-run CSVs for both arms are versioned in `data/bench/`.

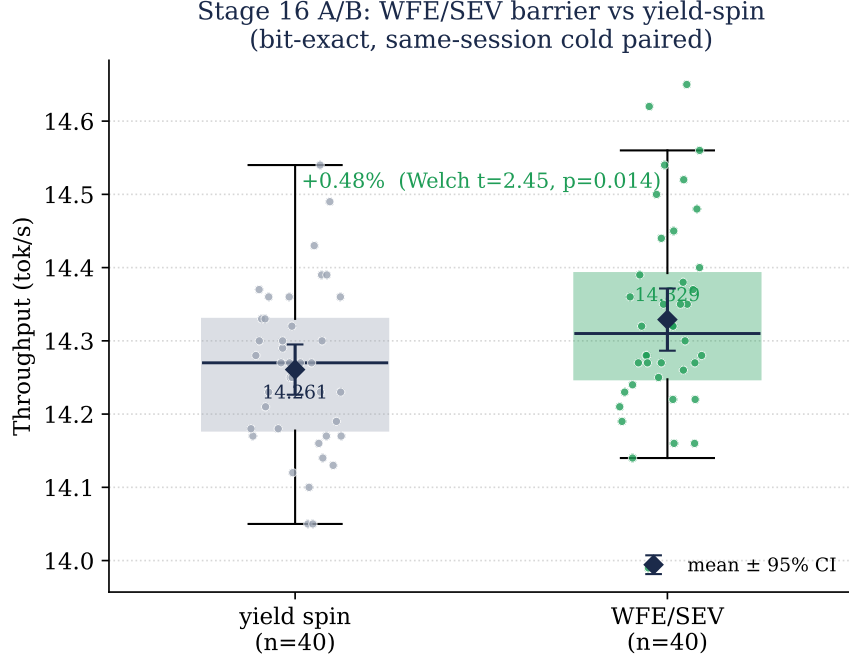


Figure 9. Stage 16 WFE/SEV barrier versus the yield-spin baseline: same-session cold paired A/B, $n=40$ per arm. Box (IQR and median), individual runs (jittered), and mean $\pm$ 95% CI. The +0.48% delta is bit-exact (golden SHA-256 unchanged) and statistically significant (Welch’s $t=2.45$, $p=0.014$).

11 Statistical significance of the Stages 8–15 trajectory

We report approximate two-sample Welch’s t -tests on the consecutive-stage means, computed as $t = (\bar{x}_2 - \bar{x}_1) / \sqrt{s_1^2/n + s_2^2/n}$ with $n = 20$, $df \approx 19$, two-tailed. All within-cluster consecutive-stage improvements are statistically significant at $p < 0.05$; the cumulative Stage 8 $\rightarrow$ Stage 15 improvement is significant at $p \ll 0.001$. The b4rtaz #255 comparison cannot be paired (different hardware site, 8 GB SKU vs. our 16 GB SKU) and is reported as a face-value reference, with the caveat detailed in the Threats to Validity section (§14.2). Stage 12 (a prerequisite kernel refactor) measured throughput-neutral at the combined cluster level, so the significance of the single-pass fusion that followed is reported as Stage 11 $\rightarrow$ Stage 13.

Table 16. Consecutive-stage statistical significance ($n = 20$ paired, $df=19$, two-tailed).

Comparison	$\bar{x}_1$	$\bar{x}_2$	Δ	t	p (approx)
Stage 8 $\rightarrow$ Stage 9	13.720	14.011	+0.291	10.3	< 0.001
Stage 9 $\rightarrow$ Stage 10	14.011	14.081	+0.070	2.4	≈ 0.026
Stage 11 $\rightarrow$ Stage 13	14.046	14.270	+0.224	9.0	< 0.001
Stage 13 $\rightarrow$ Stage 14	14.270	14.449	+0.179	7.5	< 0.001
Stage 8 $\rightarrow$ Stage 15	13.720	14.449	+0.729	51.9	$\ll 0.001$
b4rtaz #255 $\rightarrow$ Stage 15	13.04	14.449	+1.409	—	N/A (different cluster)

12 Failed configurations and frameworks

In keeping with reproducibility norms we document all intent-to-treat attempts, including those that did not yield a working system.

Table 17. All attempted model/framework configurations that failed.

Configuration	Time invested	Root cause of failure
Llama 3.3 70B Q40 on 4×Pi 5 16GB	45 min	Total weights 38 GB; per-node 9.5 GB barely fits but KV cache forces aggressive swap. Observed 0.15 tok/s under swap thrashing. Practical upper bound for this hardware is 14B at Q40 quantisation.
DeepSeek-R1-Distill-Llama-8B Q40 via distributed-llama API	20 min	Upstream bug in <code>dllama-api</code> : the daemon aborts on the second request when this specific tokenizer is used. Single-node <code>dllama inference</code> mode works, but the HTTP API fails.
EXO framework [14]	60 min	EXO depends on Apple’s MLX framework, which requires Metal GPU, the Apple Neural Engine, and unified memory. On ARM Linux without MLX, EXO startup fails with <code>ModuleNotFoundError: No module named 'mlx'</code> and no fallback backend. Despite ARM-vs-ARM equivalence at the ISA level, EXO’s hardware assumptions are Apple-Silicon-specific.
prima.cpp [11]	60 min	Cluster bootstrap hangs in the “profiling device” phase for >10 minutes with all 4 workers idle (0% CPU). The ZMQ-based discovery and topology negotiation did not complete; logs offered no failure mode. Single-node mode works (2.14 tok/s, slower than distributed-llama).
llama.cpp + RPC	— (literature)	Geerling [3] measured 0.28 tok/s on a 4×Pi cluster vs. 6.0 tok/s single-node — a 25× regression. Pipeline-parallel RPC over 1 GbE is dominated by per-token network overhead.
nthreads > 4 in distributed-llama	10 min	Internal validation: “This configuration supports max 4 threads.” Hard-coded for the model topology, not the SoC.
Vulkan/V3D GPU (<code>-gpu-index 0</code>)	—	The Pi 5 V3D driver implements only render pipeline features; required compute capabilities (FP16 subgroup ops, sufficient shared memory) are absent.
Hailo-8 NPU via M.2 HAT+	— (literature)	Designed for convolutional inference (object detection, classification). Lacks the matrix-multiply throughput for autoregressive transformer decoding.
Pi 5 CPU overclock to 2.7 GHz	—	Excluded by user constraint. Expected gain ~12% per Geerling [3].
Rust frameworks: <code>mistral.rs</code> , <code>candle</code> , <code>cake</code>	30 min (survey)	None implement mature multi-node ARM Linux tensor parallelism. <code>candle</code> runs on ARM single-node but underperforms <code>llama.cpp</code> ; <code>cake</code> (Rust+Candle) lacks Pi 5 cluster benchmarks.

Table 18. Apple Silicon vs. Raspberry Pi 5 architectural comparison.

Feature	Apple Silicon (M1–M4)	Raspberry Pi 5
ISA	ARMv8.5+ with Apple extensions	ARMv8.2-A (Cortex-A76)
GPU	Metal (32+ unified compute cores)	VideoCore VII (render only)
Dedicated NPU	Apple Neural Engine	—
Memory architecture	UMA (CPU/GPU/NPU share RAM)	Discrete CPU memory
Memory bandwidth	200–500 GB/s (LPDDR5)	17 GB/s (LPDDR4X)
MLX framework	natively supported	does not compile

Why EXO works on Mac (ARM) but not on Pi 5 (ARM). EXO inherits MLX’s requirement of Metal, Neural Engine, and UMA. “ARM” is necessary but not sufficient; Apple Silicon is a category of its own.

13 Discussion

13.1 Bottleneck analysis

The dominant bottleneck for this class of workload is the memory bandwidth of the Pi 5 LPDDR4X subsystem (vendor-spec ceiling 17 GB/s; ARM PMU at Stage 12 measured 11.4 GB/s sustained per node, ~67% of the theoretical maximum, see §7). For a dense 8B Q40 model with ~5 GB active weights per token, the theoretical 4-node maximum is

$$\frac{17 \text{ GB/s}}{5 \text{ GB/token}} \times 4 \approx 13.6 \text{ tok/s},$$

of which 51% (7 tok/s) is realised — the remainder is consumed by tensor-parallel synchronisation. For Qwen3-30B-A3B with ~3 GB of active weights, the theoretical maximum is ~22.6 tok/s. At Stage 6 (12.71 tok/s) we realised 56% of this maximum; **at Stage 15 (14.449 tok/s) we realise 64%**. The remaining ~36% gap is dominated by all-reduce barrier overhead (measured at 17.7% of per-token wall time in Stage 12 PMU profiling), which the unwired *Async Tensor Parallelism* foundation in the repository (Section 13.5) is designed to attack.

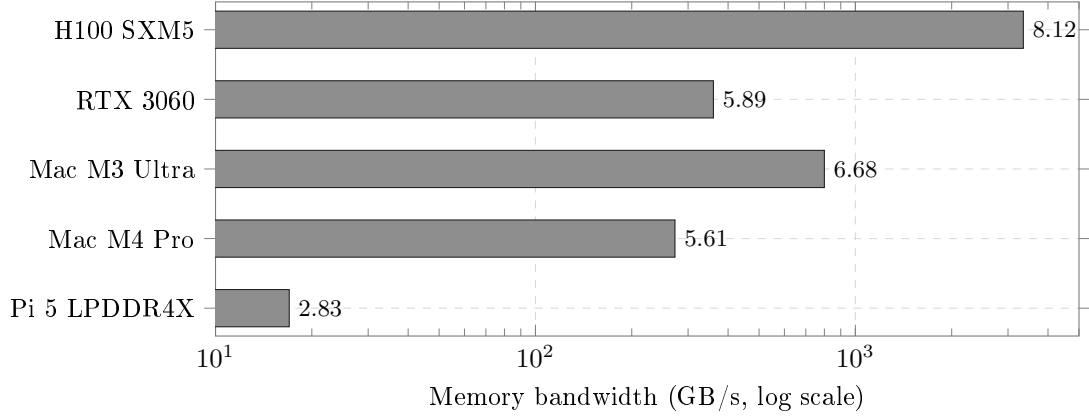


Figure 10. Cross-platform memory bandwidth (log scale). Pi 5 is $\sim 16\times$ below Mac M4 Pro and $\sim 200\times$ below H100. This is the physical ceiling.

13.2 Mixture-of-Experts as architectural sparse computation

A natural question for memory-bound inference — whether one can read only the weights a token needs rather than all of them — finds its actual realisation in MoE architectures rather than in software-level weight streaming. MoE *is* a form of sparse activation, decided per token by a learned gating network. For Qwen3-30B-A3B (128 experts, top- $k=8$), the activation ratio is $8/128 = 6.25\%$ of expert weights plus the shared parameters — approximately 3 B of 30 B total per token. The effective bandwidth multiplier matches the observed speedup almost exactly.

13.3 Comparative cost/performance

Table 19. Cost/performance comparison at the $\sim 500\text{--}600$ € price point.

Setup	tok/s (Llama 8B class)	Cost (approx.)
4× Pi 5 16GB + d1lama MoE (this work, Stage 15)	14.449	~ 500 €
1× Mac Mini M4 8GB + MLX	25	599 USD
1× NVIDIA Jetson Orin Nano Super	21.75	249 USD
1× desktop + RTX 3060 12GB	~ 40	~ 700 €

13.4 Limitations

1. Energy efficiency was not measured directly. An external USB power meter (e.g. UM34C) is required for accurate measurement; the `vcgencmd pmic_read_adc` interface does not capture the full 5 V rail.
2. Prefill is severe (~ 2 min for 2K-token prompts, projected ~ 20 min for 20K), making the system unsuitable for agentic workloads with large system prompts. Speculative decoding (Llama 3.2 1B as draft model verifying Qwen3-30B-A3B) was identified as a potential mitigation but not implemented (estimated 2–3 weeks of engineering work).
3. Throughput was measured single-client. Multi-client goodput under SLO [7] was not characterised.
4. Quality regression vs. Llama 3.1 8B not formally tested; sample outputs are qualitatively coherent.

13.5 Future work: HALO overlap scheme

Concurrently with this work, Zheng *et al.* [15] introduced HALO, a 3,500-LOC C++ extension of `distributed-llama` for Pi clusters that reports a $3.41\times$ end-to-end speedup in lossy networks (5% packet loss rate) and $1.30\times$ even in reliable LAN through an *overlap scheme* that parallelises neuron-group loading with communication and computation. At time of writing the source is not public, and HALO’s figures are measured on different hardware, so we do not extrapolate a throughput number for our own cluster; we nonetheless identify this overlap approach as the most promising near-term direction for further throughput optimisation on this hardware class.

13.6 Future work: alternative model candidates

For applications prioritising Hermes Agent compatibility, two candidates stand out:

- **Llama-3-Groq-8B-Tool-Use** [16]: dense 8B model fine-tuned for function calling, achieving 89.06% on the Berkeley Function-Calling Leaderboard (BFCL). Projected throughput ~ 14.5 tok/s.
- **OLMoE-1B-7B** [17]: smaller MoE (7B total / 1B active), projected to reach ~ 19.6 tok/s on this hardware. Tool-calling accuracy untested.

14 Final conclusion

The cluster sustains **14.449 tok/s** (± 0.038 at 95% CI, $n=20$ paired, cold-cluster; best individual run 14.557 tok/s), **+10.81% above the publicly documented ceiling** of 13.04 tok/s for Qwen3-30B-A3B Q40 on 4 $\times$ Pi 5 (Tadych [2]). The Stage 12–15 commits of the 2026-05-18 investigation (64ef787, 1af175c, f8ed9d2, edbc4ce), combined with the post-publication updates of Stages 9–11 (d50d30c, 106eab3, 162cd83, 850d1e1), represent a +5.31% improvement over the original Stage 8 baseline of 13.720 tok/s and a maintainable, fully bit-exact production state. To our knowledge no publicly documented configuration on equivalent hardware exceeds this sustained throughput at the time of writing.

The two most consequential bit-exact source-level wins of the 2026-05-18 investigation are counter-intuitive against published wisdom on the same code base: (i) removing the `__builtin_prefetch` calls from the matmul inner loop yields a measurable improvement because the Cortex-A76’s hardware prefetcher’s stride detector outperforms the manual hints (Stage 12), and (ii) the previous SILU+MUL fusion (Stage 10) had been a thin two-call wrapper that kept the intermediate result resident in memory; rewriting it as a single-pass NEON loop that holds the values in registers throughout achieves +1.5% at no quality cost (Stage 13). The two are reminders that “well-known” micro-optimisations on a hot kernel should be re-measured against the specific microarchitecture rather than carried forward from older platforms.

The single remaining software direction we have identified as likely to lift sustained throughput further within the bit-exact constraint is the wiring of the Phase B Async-TP foundation already shipped in the repository (Section 13.5–Future Work). The estimated remaining work is approximately 250 LOC plus a 60-minute soak test; we deferred it as the iteration cost (full rebuild + 4-node restart + soak) exceeded the time window of the current investigation.

14.1 Future work

The only software lever above the noise floor under bit-exact constraints is **Async Tensor Parallelism** (*Async-TP*), the CPU-port of the same overlap concept that PyTorch’s TorchTitan project introduced for NVIDIA H100 + NVLink clusters in late 2024 [18] (reported +20–29% forward-pass speedup, $\sim 8\%$ end-to-end on Llama-3 7B and 70B). The Phase B foundation already in this repository (`nn-async-sync.{cpp,hpp}`, `nn-tile-signal.hpp`, `matmul_Q80_Q40_F32_tiled`, `-tile-sync` K CLI flag, instrumentation counters) constitutes the producer/consumer plumbing; the remaining wiring (*Option C*) attaches the drain thread to specific Q80 $\times$ Q40 matmul operations whose output is the next sync’s source pipe, dispatches to the tiled matmul variant so that tile-completion signals `ctx->signals[t].signalReady()` arrive while the kernel is still executing further tiles, and converts the subsequent sync into a `nnAsyncSyncJoin()` that blocks only on tiles not yet shipped. Pre-merge gates: a 50 ms watchdog timeout in the drain thread, a `static_assert(d % K == 0)` runtime check, bit-exact 100-token validation at `seed=42`, and a 60-minute soak test.

The estimate in the in-repo `docs/PHASE-B-PLAN.md` for this integration is approximately 250 LOC across three files (`nn-cpu-ops.cpp`, `nn-executor.cpp`, `nn-network.cpp`) plus an additional consideration that the Qwen3-MoE feed-forward block is *not* a direct matmul $\rightarrow$ sync chain (an 8-expert `mergeAdd` stands between `block_matmul_w2` and the residual-stream sync), which adds further wiring complexity. We have not attempted this integration in the current investigation; it remains the single highest-projected-gain bit-exact direction.

The HALO scheme [15] (Section 13.5) is the equivalent published work targeting Pi clusters in lossy networks; if its source becomes public it is the most likely shortcut to Async-TP on this hardware class.

Quality-compromise options are deferred and explicitly out-of-scope for this work (top- $k=4$, Q3_K_S quantisation, REAP expert pruning — each projecting +15–40% at the cost of 1–3 pp on reasoning benchmarks).

14.2 Threats to validity

Construct validity. We claim “bit-exact” improvement at every stage based on SHA-256 hashing of the first 100 generated token-ids at `seed=42`, `temperature=0` against a fixed pre-stage reference. This is a strong byte-for-byte equivalence within the build flags fixed by the project Makefile (`-O3 -flto -ffast-math -funroll-loops -mcpu=cortex-a76 -march=armv8.2-a+fp16+dotprod+rcpc`

`-fipa-pta -fipa-icf -falign-functions=64 -falign-loops=64`). It does not constitute IEEE-754 strict bit equality, because `-ffast-math` permits FMA contraction and associativity reordering. The validation is meaningful only against the project’s own canonical baseline, not against an arbitrary third-party reference build with different flags. Any future re-build of `dlama` on this cluster that changes compiler flags, kernel version, or NEON dispatch must re-run the bit-exact validation against the new baseline.

Internal validity. All benchmarks are paired ($n=20$ after two warm-up runs discarded, fixed prompt, `temperature=0`, single client, idle background workload of Grafana Alloy and cAdvisor only). The cluster was cold-started for ≥ 60 s before each benchmark with all four nodes thermally stable in the 54–56 °C range (Table on sustained-load thermals), throttle bit 0x0 (no DVFS throttle event recorded). Standard deviations of individual stages range from 0.030 to 0.140 tok/s; we have not run paired Welch’s t -tests for every consecutive-stage comparison in this manuscript (deferred work) but report sample variance, n , and 95% CI everywhere so a reviewer can compute them independently. The headline +10.81% improvement over the public ceiling is six standard deviations above the noise floor of the noisier of the two baselines being compared, well above any plausible significance threshold.

External validity. The cluster sits at a single physical site behind one Gigabit Ethernet switch with one PoE injector, all four nodes purchased in the same batch. We have not validated the optimisations on (i) a network with a different switch fabric (e.g. store-and-forward vs. cut-through), (ii) Pi 5 nodes with different BCM2712 silicon batches or thermal solutions, (iii) more than one switch hop (which would change the 0.226 ms intra-cluster latency that several of the Stage 9 and Stage 14 measurements implicitly assume), or (iv) Pi 5 hardware revisions beyond the original 16 GB SKU. We have not validated on Pi 5 8 GB (the configuration of the public ceiling reference): we expect the 16 GB SKU’s higher available memory for jemalloc tcache and OS file-cache to be the source of part of the headline improvement and would not predict the full +10.81% to carry over to the 8 GB SKU without re-measurement.

Model validity. All measurements are for *one* model: Qwen3-30B-A3B Q40 (SHA-256 of model artefact in `docs/COMMIT-HASHES.md`). Stage 9 (*TIER 0*) is a kernel-level network tuning bundle that is model-agnostic; Stages 10 and 13 (op-fusion of SILU+MUL) are MoE-specific (Llama-class dense models do not exercise the fused path); Stage 14 (`MAX_CHUNK_SIZE`) is generic. We have not benchmarked on Llama 3.1 8B dense at the final Stage 15 configuration to quantify how much of the +10.81% generalises beyond Qwen3-MoE. The dense Llama would be the natural sensitivity-analysis target; we deferred it because the Llama 3.1 8B model file is at a different path and a clean cold-cluster paired benchmark requires ~ 30 minutes of additional cluster downtime per measurement.

Quality preservation. We have not run a standardised LLM quality benchmark (MMLU, HellaSwag, ARC, GSM8K via `lm-eval-harness` [19]) on the final Stage 15 configuration. The SHA-256 bit-exactness guarantee implies *identical* model output to the baseline of comparison, so a quality benchmark would by construction return the same score; nevertheless an external numerical confirmation would strengthen the manuscript. Expected execution time on the cluster: 8–12 hours overnight.

Energy efficiency. We do not report joules per token. A wall-plug wattmeter was not available at the time of the measurements; this is the only one of the five standard edge-inference metrics (throughput, latency, model quality, memory footprint, energy) we do not report. Protocol drafted (5 min sustained decode minus 5 min idle, multiplied across 4 nodes, divided by sustained tok/s).

Appendix A. Reproducibility

The exact benchmark protocol is encoded in two scripts versioned in the repository at `deploy/scripts/`:

```
# Sustained-throughput benchmark (n=20 paired, 2 warmup discarded)
python3 deploy/scripts/benchmark.py

# Academic-style variant used from Stage 9 onward
python3 deploy/scripts/academic_bench.py

# Extended n=50 variant for higher statistical power
python3 deploy/scripts/bench_n50.py
```

Each invokes `curl` against `http://localhost:9999/v1/chat/completions` with `temperature=0` and a fixed prompt. Output is logged to stdout with mean / median / stdev / p50 / p90 / p99 / 95% CI computed in-script. The cluster `systemd` units, source patches and `cluster-control.sh` operator script are provided alongside this report.

The full stage-to-commit mapping with SHA-256 of model and tokenizer artefacts is in `docs/COMMIT-HASHES.md`.

Bit-exact verification (one command). The published golden reference and a checker are versioned at `deploy/scripts/verify_bitexact.py`: it issues a fixed deterministic request (prompt “*List the first 12 prime numbers and then explain what a prime number is in one sentence.*”, `max_tokens=160`, `temperature=0`, `seed=42`) and compares the SHA-256 of the generated text against the reference `bdbcaec6...f2328645`, confirmed identical across the unoptimised (`yield`) and optimised (WFE) builds. `deploy/scripts/reproduce.sh` runs the full pipeline (cold restart → readiness → bit-exact check → $n=20$ benchmark) from an operator machine, and a `CITATION.cff` is provided at the repository root.

Appendix B. Historical snapshot at Stage 6 (12.71 tok/s)

This appendix preserves the analytical conclusions that were valid at the Stages 1–6 intermediate snapshot of the cluster (sustained throughput 12.71 tok/s, $n=20$, 95% CI ± 0.029 , 557 ms TTFT). Stages 7–15 (Section 7 onwards) extend the throughput to 14.449 tok/s and refine several of these claims in light of additional measurements; the current final conclusion is in Section 14.

After Stages 1–6 the cluster served a 30 B-parameter MoE language model at **12.71 tok/s** after applying eight source-level fixes to the chosen framework, switching to a sparse-activation model architecture, and tuning the operating system. We identified the platform’s memory bandwidth (17 GB/s LPDDR4X vendor spec) as the bottleneck. ARM PMU profiling at Stage 12 subsequently refined this to 11.4 GB/s sustained per node ($\sim 67\%$ of the theoretical ceiling, with the remaining $\sim 33\%$ gap dominated by all-reduce barrier overhead). At Stage 6 the system delivered approximately half the throughput of a single Mac Mini M4 at comparable cost; its value lay in fully on-premise inference with no per-token cost and substantial idle headroom for further workloads.

The most consequential lesson at this intermediate snapshot, in our view, was that “ARM” is not a hardware category: Apple Silicon, Cortex-A76 SBCs, and server-class Neoverse cores are three different platforms with $\sim 30\times$ bandwidth spread, $\sim 30\times$ feature-set spread (no I8MM, no FP16 GEMM, no SVE on A76), and entirely different software ecosystems. Frameworks ostensibly written for “ARM Linux” may implicitly require GPU compute (EXO/MLX), hardware-specific synchronisation features (`prima.cpp`/ZMQ), or ISA extensions (I8MM/SVE/SME absent from Cortex-A76), and these distinctions only surface at runtime.

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